

5075. Big Data aplicado - 1ª Evaluación (RA 1 – CE c)

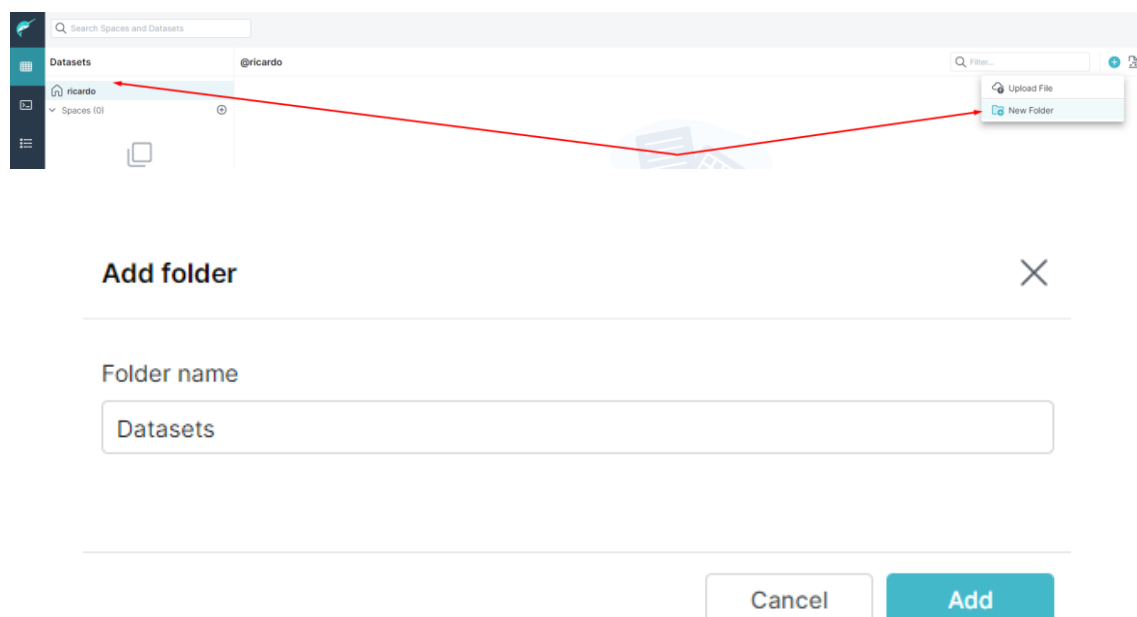
Unidad Didáctica 1. Estrategias de ingestión y almacenamiento de datos en Big Data

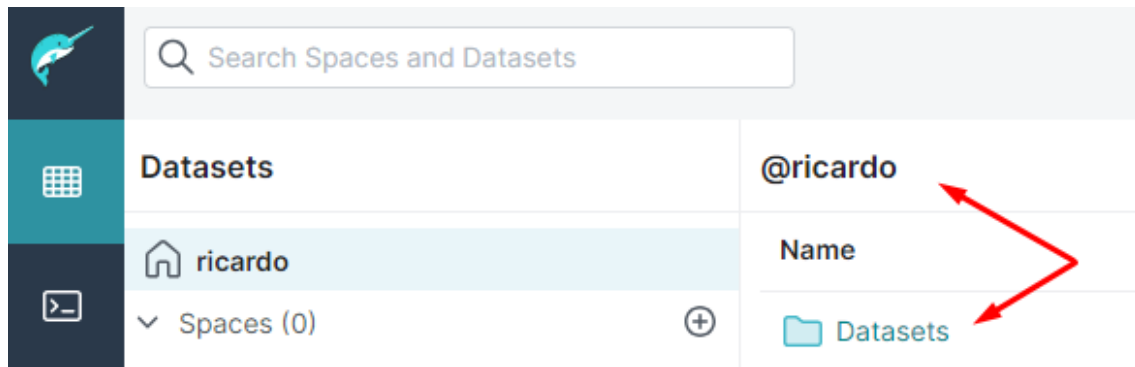
Práctica 9. Trabajando con distintos formatos en Dremio.

Paso 1. Nos descargamos el fichero CSV del enlace https://raw.githubusercontent.com/ricardomariscalquintero/bda/main/supermarket_sales.csv.

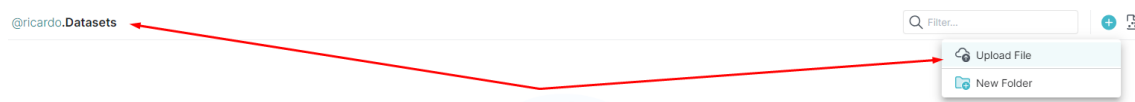
```
Invoice ID,Branch,City,Customer type,Gender,Product line,Unit price,Quantity,Tax 5%,Total,Date,Time,Payment,cogs,gross margin percentage,gross income,Rating
750-67-8428,A,Yangon,Member,Female,Health and beauty,74.69,7,26.1415,548.9715,1/5/2019,13:08,Ewallet,522.83,4.761904762,26.1415,9.1
226-31-3081,C,Naypyitaw,Normal,Female,Electronic accessories,15.28,5,3.82,80.22,3/8/2019,10:29,Cash,76.4,4.761904762,3.82,9.6
631-41-3108,A,Yangon,Normal,Male,Home and lifestyle,46.33,7,16.2155,340.5255,3/3/2019,13:23,Credit card,324.31,4.761904762,16.2155,7.4
123-19-1176,A,Yangon,Member,Male,Health and beauty,58.22,8,23.288,489.048,1/27/2019,20:33,Ewallet,465.76,4.761904762,23.288,8.4
373-73-7910,A,Yangon,Normal,Male,Sports and travel,86.31,7,30.2085,634.3785,2/8/2019,10:37,Ewallet,604.17,4.761904762,30.2085,5.3
699-14-3026,C,Naypyitaw,Normal,Male,Electronic accessories,85.39,7,29.8865,627.6165,3/25/2019,18:30,Ewallet,597.73,4.761904762,29.8865,4.1
355-53-5943,A,Yangon,Member,Female,Electronic accessories,68.84,6,20.652,433.692,2/25/2019,14:36,Ewallet,413.04,4.761904762,20.652,5.8
315-22-5665,C,Naypyitaw,Normal,Female,Home and lifestyle,73.56,10,36.78,772.38,2/24/2019,11:38,Ewallet,735.6,4.761904762,36.78,8
665-32-9167,A,Yangon,Member,Female,Health and beauty,36.26,2,3.626,76.146,1/10/2019,17:15,Credit card,72.52,4.761904762,3.626,7.2
692-92-5582,B,Mandalay,Member,Female,Food and beverages,54.84,3,8.226,172.746,2/20/2019,13:27,Credit card,164.52,4.761904762,8.226,5.9
351-62-0022,B,Mandalay,Member,Female,Fashion accessories,14.48,4,2.896,60.816,2/6/2019,18:07,Ewallet,57.92,4.761904762,2.896,4.5
529-56-3974,B,Mandalay,Member,Male,Electronic accessories,25.51,4,5.102,107.142,3/9/2019,17:03,Cash,102.04,4.761904762,5.102,6.8
365-64-0515,A,Yangon,Normal,Female,Electronic accessories,46.95,5,11.7375,246.4875,2/12/2019,10:25,Ewallet,234.75,4.761904762,11.7375,7.1
252-56-2699,A,Yangon,Normal,Male,Food and beverages,43.19,10,21.595,453.495,2/7/2019,16:48,Ewallet,431.9,4.761904762,21.595,8.2
829-34-3910,A,Yangon,Normal,Female,Health and beauty,71.38,10,35.69,749.49,3/29/2019,19:21,Cash,713.8,4.761904762,35.69,5.7
299-46-1805,B,Mandalay,Member,Female,Sports and travel,93.72,6,28.116,590.436,1/15/2019,16:19,Cash,562.32,4.761904762,28.116,4.5
656-95-9349,A,Yangon,Member,Female,Health and beauty,68.93,7,24.1255,506.6355,3/11/2019,11:03,Credit card,482.51,4.761904762,24.1255,4.6
765-26-6951,A,Yangon,Normal,Male,Sports and travel,72.61,6,21.783,457.443,1/1/2019,10:39,Credit card,435.66,4.761904762,21.783,6.9
329-62-1586,A,Yangon,Normal,Male,Food and beverages,54.67,3,8.2005,172.2105,1/21/2019,18:00,Credit card,164.01,4.761904762,8.2005,8.6
319-50-3348,B,Mandalay,Normal,Female,Home and lifestyle,40.3,2,4.03,84.63,3/11/2019,15:30,Ewallet,80.6,4.761904762,4.03,4.4
300-71-4605,C,Naypyitaw,Member,Male,Electronic accessories,86.04,5,21.51,451.71,2/25/2019,11:24,Ewallet,430.2,4.761904762,21.51,4.8
371-85-5789,B,Mandalay,Normal,Male,Health and beauty,87.98,3,13.197,277.137,3/5/2019,10:40,Ewallet,263.94,4.761904762,13.197,5.1
273-16-6619,B,Mandalay,Normal,Male,Home and lifestyle,33.2,2,3.32,69.72,3/15/2019,12:20,Credit card,66.4,4.761904762,3.32,4.4
636-48-8204,A,Yangon,Normal,Male,Electronic accessories,34.56,5,8.64,181.44,2/17/2019,11:15,Ewallet,172.8,4.761904762,8.64,9.9
549-59-1358,A,Yangon,Member,Male,Sports and travel,88.63,3,13.2945,279.1845,3/2/2019,17:36,Ewallet,265.89,4.761904762,13.2945,6
```

Paso 2. Creamos una carpeta para albergar los ficheros.

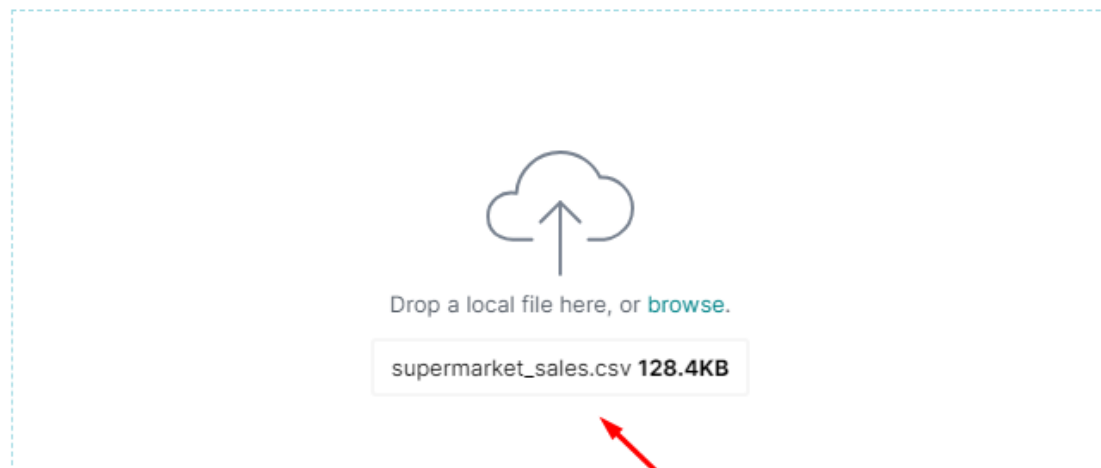




Paso 3. Cargamos el archivo.



Add File: Browse for File (Step 1 of 2)



Name

Supermarket

Cancel

Next

Add File: Set Format (Step 2 of 2)

Format

Text (delimited)

Column Delimiter

Comma

Quote

Double Quote

Comment

Number Sign

Line Delimiter

CRLF - Windows

Escape

Double Quote

Options

☒ Extract Column Names ☐ Skip First Line
☒ Trim Column Names

abc Invoice ID	abc Branch	abc City	abc Customer type	abc Gender	abc Product line
750-67-8428	A	Yangon	Member	Female	Health and beauty
226-31-3081	C	Naypyitaw	Normal	Female	Electronic accessories
631-41-3108	A	Yangon	Normal	Male	Home and lifestyle
123-19-1176	A	Yangon	Member	Male	Health and beauty
373-73-7910	A	Yangon	Normal	Male	Sports and travel

 Search Spaces and Datasets

Datasets

 ricardo

Spaces (0)

@ricardo.Datasets

Name

 Supermarket

Run Preview

Hide SQL pane Save as View

Context: (None selected)

1 SELECT * FROM "ricardo".Datasets.Supermarket

Query1

Add Column Group By Join Filter Columns 17 Columns Job: Run Rows: 1,000 <1s

abc Invoice ID	abc Branch	abc City	abc Customer type	abc Gender	abc Product line
750-67-8428	A	Yangon	Member	Female	Health and beauty
226-31-3081	C	Naypyitaw	Normal	Female	Electronic accessories
631-41-3108	A	Yangon	Normal	Male	Home and lifestyle
123-19-1176	A	Yangon	Member	Male	Health and beauty
373-73-7910	A	Yangon	Normal	Male	Sports and travel
699-14-3026	C	Naypyitaw	Normal	Male	Electronic accessories
355-53-5943	A	Yangon	Member	Female	Electronic accessories
315-22-5665	C	Naypyitaw	Normal	Female	Home and lifestyle
665-32-9167	A	Yangon	Member	Female	Health and beauty
692-92-5582	B	Mandalay	Member	Female	Food and beverages
351-62-0822	B	Mandalay	Member	Female	Fashion accessories
529-56-3974	B	Mandalay	Member	Male	Electronic accessories
365-64-0515	A	Yangon	Normal	Female	Electronic accessories

Job: Run Rows: 1,000 <1s. Download (sample) (i)

JSON

CSV

Parquet

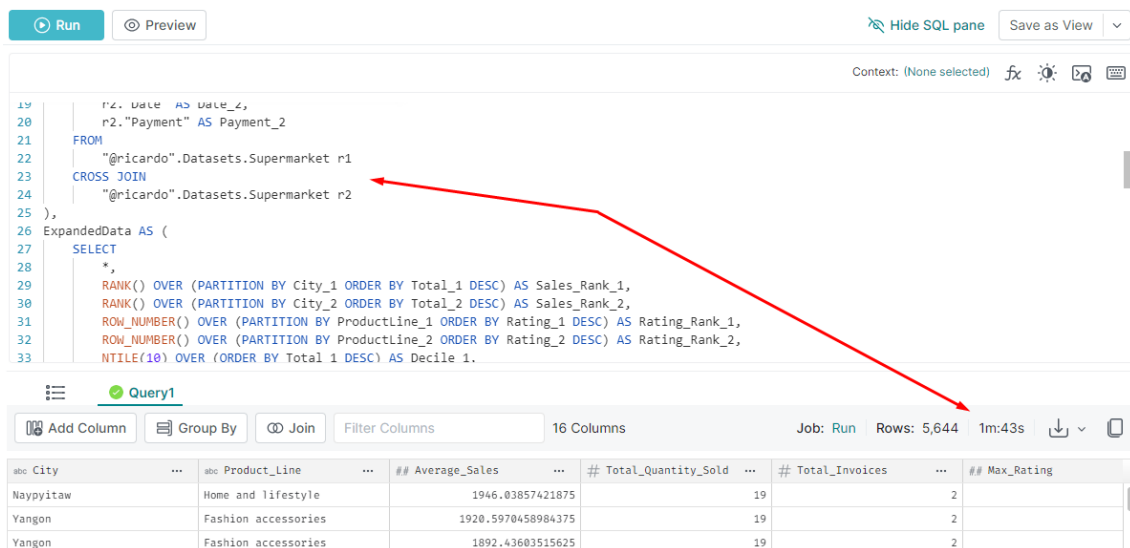
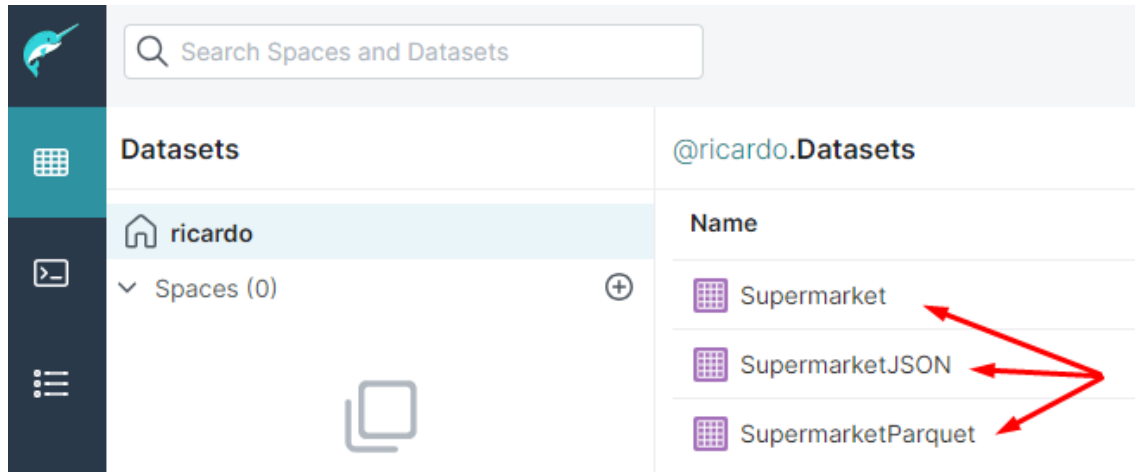
...	abc Gender
	Female
	Female
	Male
	Male
	Male

 supermarket_sales	24/10/2019 11:07	Archivo de valores...	129 KB
 supermarket_sales.json	04/09/2024 21:55	Archivo JSON	349 KB
 supermarket_sales.parquet	04/09/2024 21:55	Archivo PARQUET	53 KB

1	{ "Invoice ID": "7500-67-0428", "Branch": "A", "City": "Yangon", "Customer type": "Member", "Gender": "Female", "Product line": "Health and beauty", "Unit price": 74
2	{ "Invoice ID": "226-31-8081", "Branch": "C", "City": "Naypyitaw", "Customer type": "Normal", "Gender": "Female", "Product line": "Electronic accessories", "Unit price": 91
3	{ "Invoice ID": "631-41-3108", "Branch": "A", "City": "Yangon", "Customer type": "Normal", "Gender": "Male", "Product line": "Home and lifestyle", "Unit price": 44
4	{ "Invoice ID": "123-19-1176", "Branch": "A", "City": "Yangon", "Customer type": "Member", "Gender": "Male", "Product line": "Health and beauty", "Unit price": 58.2
5	{ "Invoice ID": "373-73-7910", "Branch": "A", "City": "Yangon", "Customer type": "Normal", "Gender": "Male", "Product line": "Sports and travel", "Unit price": 86.3
6	{ "Invoice ID": "699-14-3026", "Branch": "C", "City": "Naypyitaw", "Customer type": "Normal", "Gender": "Male", "Product line": "Electronic accessories", "Unit price": 91
7	{ "Invoice ID": "355-53-5943", "Branch": "A", "City": "Yangon", "Customer type": "Member", "Gender": "Female", "Product line": "Electronic accessories", "Unit price": 86.3
8	{ "Invoice ID": "315-22-5665", "Branch": "C", "City": "Naypyitaw", "Customer type": "Normal", "Gender": "Female", "Product line": "Home and lifestyle", "Unit price": 44
9	{ "Invoice ID": "665-32-9167", "Branch": "A", "City": "Yangon", "Customer type": "Member", "Gender": "Female", "Product line": "Health and beauty", "Unit price": 58.2
10	{ "Invoice ID": "692-92-5582", "Branch": "B", "City": "Mandalay", "Customer type": "Member", "Gender": "Female", "Product line": "Food and beverages", "Unit price": 43
11	{ "Invoice ID": "351-62-0822", "Branch": "B", "City": "Mandalay", "Customer type": "Member", "Gender": "Female", "Product line": "Fashion accessories", "Unit price": 86.3
12	{ "Invoice ID": "365-63-0822", "Branch": "B", "City": "Mandalay", "Customer type": "Member", "Gender": "Male", "Product line": "Electronics", "Unit price": 86.3
13	{ "Invoice ID": "365-64-0515", "Branch": "A", "City": "Yangon", "Customer type": "Normal", "Gender": "Female", "Product line": "Electronic accessories", "Unit price": 91
14	{ "Invoice ID": "252-56-2699", "Branch": "A", "City": "Yangon", "Customer type": "Normal", "Gender": "Male", "Product line": "Food and beverages", "Unit price": 43
15	{ "Invoice ID": "829-34-3910", "Branch": "A", "City": "Yangon", "Customer type": "Normal", "Gender": "Female", "Product line": "Health and beauty", "Unit price": 58.2

[illegible]

Paso 5. Subimos los archivos y lanzamos una consulta ineficiente. Vamos a comprobar sobre que dataset tarda menos tiempo.



```
19  r2.Date AS Date_2,
20  r2.Payment AS Payment_2
21  FROM
22  "@ricardo".Datasets.Supermarket r1
23  CROSS JOIN
24  "@ricardo".Datasets.Supermarket r2
25 ),
26 ExpandedData AS (
27  SELECT
28  *,
29  RANK() OVER (PARTITION BY City_1 ORDER BY Total_1 DESC) AS Sales_Rank_1,
30  RANK() OVER (PARTITION BY City_2 ORDER BY Total_2 DESC) AS Sales_Rank_2,
31  ROW_NUMBER() OVER (PARTITION BY ProductLine_1 ORDER BY Rating_1 DESC) AS Rating_Rank_1,
32  ROW_NUMBER() OVER (PARTITION BY ProductLine_2 ORDER BY Rating_2 DESC) AS Rating_Rank_2,
33  NTILE(10) OVER (ORDER BY Total_1 DESC) AS Decile_1.
```

City	Product_Line	Average_Sales	Total_Quantity_Sold	Total_Invoices	Max_Rating
Naypyitaw	Home and lifestyle	1946.03857421875	19	2	
Yangon	Fashion accessories	1920.5970458984375	19	2	
Yangon	Fashion accessories	1892.43603515625	19	2	

 **Supermarket**

Job: Run Rows: 5,644 1m:43s

 SupermarketJSON

Job: Run Rows: 5,614 1m:08s  

 SupermarketParquet

Job: Run Rows: 5,629 1m:04s  

Paso 6. Revisamos los detalles de los trabajos.



Search Jobs

Job ID

✓ ...a5-d144-4fc9-7cb392143d00

✓ ...c9-3204-d93a-024d9d9f8e00

✓ ...7f73-13dc-a0fb-ca28fe336100

✗ ...a057-c01b-f247-f8f81434b100

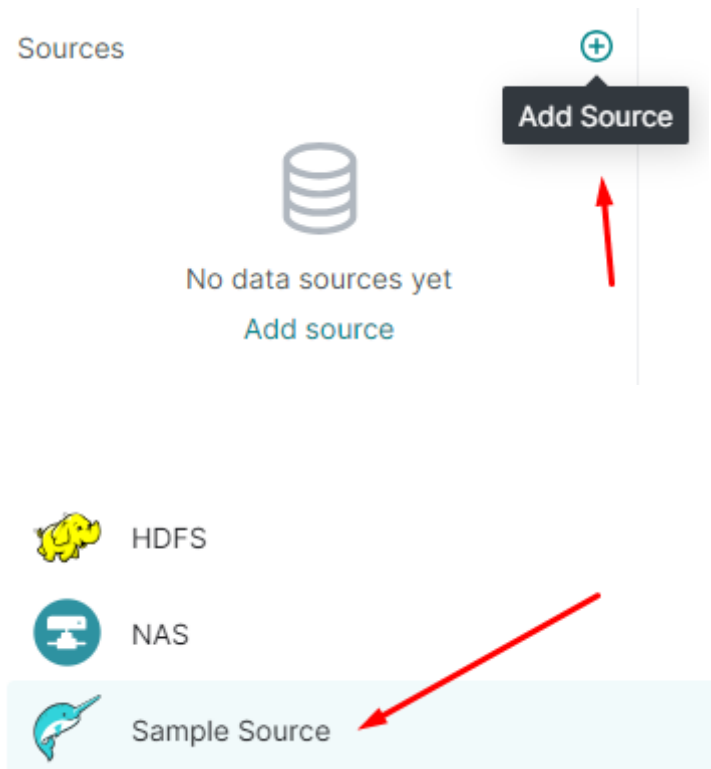
✓ ...1a-02c5-9394-0487bdedaa00

Job ID	User	 Dataset	Query Type	Queue	Start Time	Duration
✓ ...a5-d144-4fc9-7cb392143d00	ricardo	 SupermarketParquet	UI (run)	LARGE	09/04/2024, 22:05:07	00:01:04 
✓ ...c9-3204-d93a-024d9d9f8e00	ricardo	 SupermarketJSON	UI (run)	LARGE	09/04/2024, 22:03:25	00:01:08 
✓ ...7f73-13dc-a0fb-ca28fe336100	ricardo	 Supermarket	UI (run)	LARGE	09/04/2024, 21:59:39	00:01:43 

Summary

Status:	COMPLETED
Total Memory:	4.27 GB
CPU Used:	02m:37s
Query Type:	UI (run)
Start Time:	04/09/2024 22:05:07
Duration:	01m:04s
Wait on Client:	<1s
User:	ricardo
Queue:	LARGE
Input:	283.27 KB / 6K Rows
Output:	742.93 KB / 5.6K Rows

Paso 7. Comprobamos que el formato Parquet es autodescriptivo. Esto lo podremos saber mirando si los campos aparecen con sus tipos de datos.









Datasets

 ricardo

> Spaces (0) 

Sources 

Object Storage (1)

 Samples 0

Samples

Name

 samples.dremio.com



-  NYC-taxi-trips-iceberg
 -  NYC-taxi-trips.csv
 -  NYC-weather.csv
 -  SF weather 2018-2019.csv
 -  SF_incidents2016.json
 -  tpcds_sf1000
 -  zip_lookup.csv
 -  zips.json
- 

Line Delimiter

CRLF - Windows

Escape

\r\n

Double Quote



abc A	abc B	abc C
STATION	NAME	LATITUDE
USW00023272	SAN FRANCISCO DOWNTOWN, CA US	37.7705
USW00023272	SAN FRANCISCO DOWNTOWN, CA US	37.7705

Table Settings for SF_incidents2016.json

Format

Format

JSON



abc IncidentNum	abc Category	abc Descript
120058272	WEAPON LAWS	POSS OF PROHIBITED WEAPON
120058272	WEAPON LAWS	FIREARM, LOADED, IN VEHICLE,

[Samples.samples.dremio.com](https://samples.samples.dremio.com).Dremio University

Name

 100_Sales_Records_inconsistency.csv

 4week_recipes.json

 aac_shelter_outcomes.csv

 airbnb_listings.csv

 employees.parquet

 googleplaystore.csv

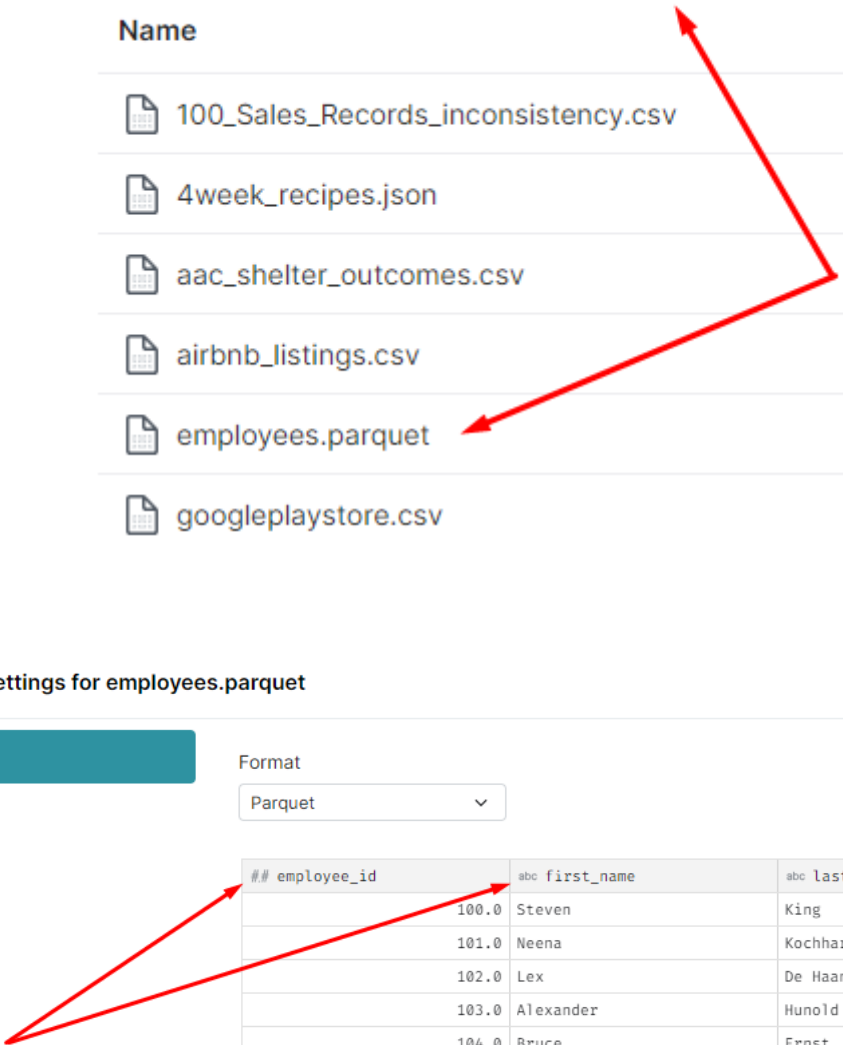
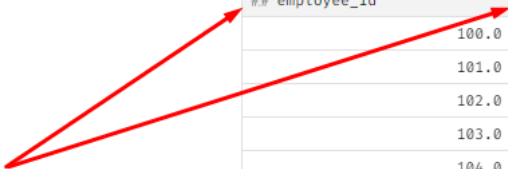


Table Settings for employees.parquet

Format

Format

Parquet



## employee_id	abc first_name	abc last_name
100.0	Steven	King
101.0	Neena	Kochhar
102.0	Lex	De Haan
103.0	Alexander	Hunold
104.0	Bruce	Ernst
105.0	David	Austin
106.0	Valli	Pataballa
107.0	Diana	Lorentz
108.0	Nancy	Greenberg
109.0	Daniel	Faviet

Como hemos podido comprobar, mientras que en CSV y en JSON todos los campos eran de tipo texto, en Parquet había distintos tipos de datos como cadenas o fechas.